OENG1209: Engineering Science

Week 7:

Momentum & Impulse

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Outline

- Momentum and Impulse
- Conservation of Momentum
- Collisions

Solve the collision

- Momentum in two dimensions

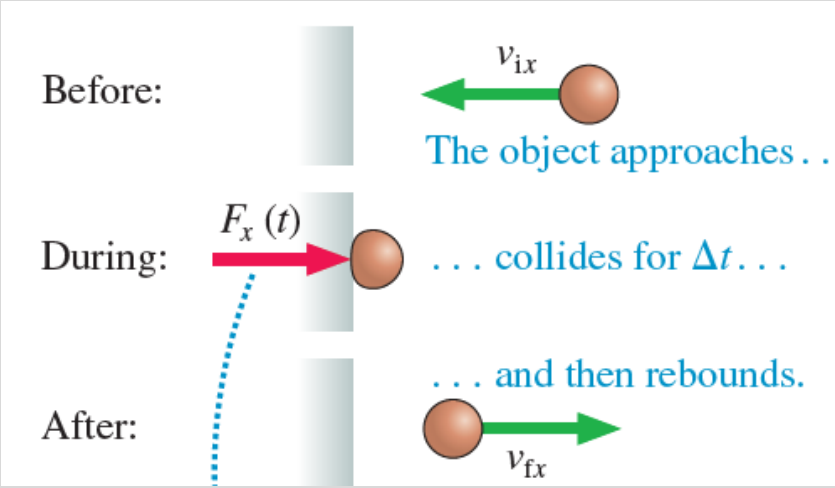
1. Momentum

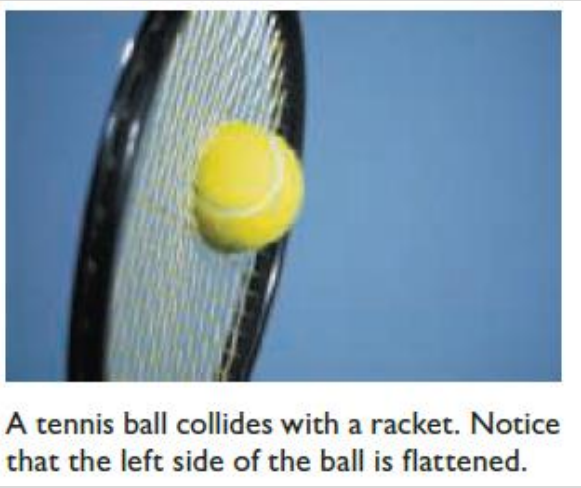
anyone or anything that has a lot of momentum is going to be hard to stop

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Momentum
Introduction

A collision is a **short-duration** interaction between two objects. Let's examine a rubber ball colliding with a wall. can't be a particle → deform object

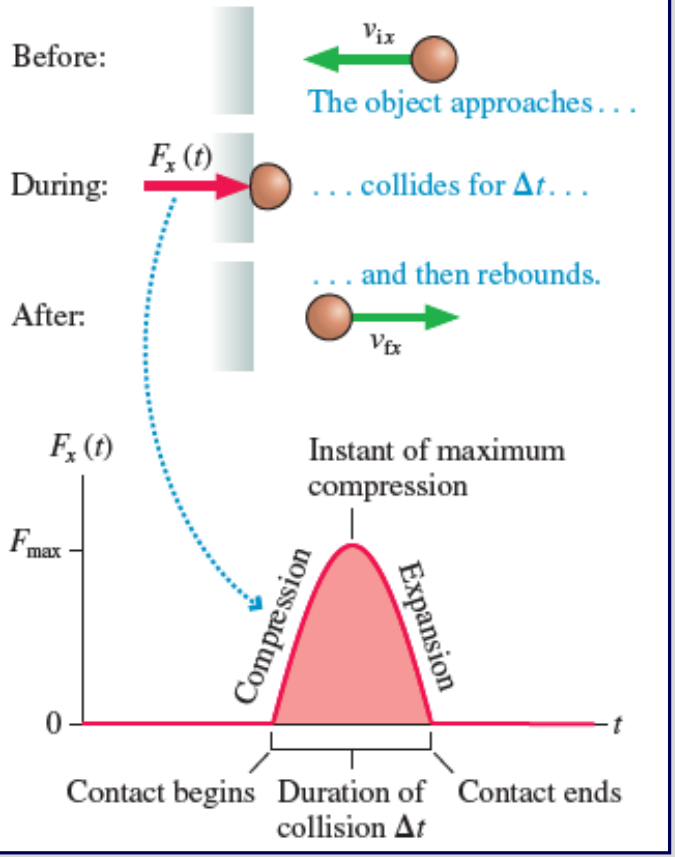




A tennis ball collides with a racket. Notice that the left side of the ball is flattened.

The force of a collision → **Impulsive force** (very **large** force exerting over a very **small** interval of time).

Momentum
Impulsive force



Goal: Given m, v_i and $F(t)$ → v_f?

use Newton's 2nd Law: $ma = F$ (not constant)

$\Rightarrow m\frac{dv}{dt} = F(t) \Rightarrow m\,dv = F(t)\,dt \Rightarrow \int m\,dv = \int F(t)\,dt \Rightarrow mv\Big|_{v_i}^{v_f} = \int F(t)\,dt \Rightarrow m(v_f - v_i) = \int F(t)\,dt$

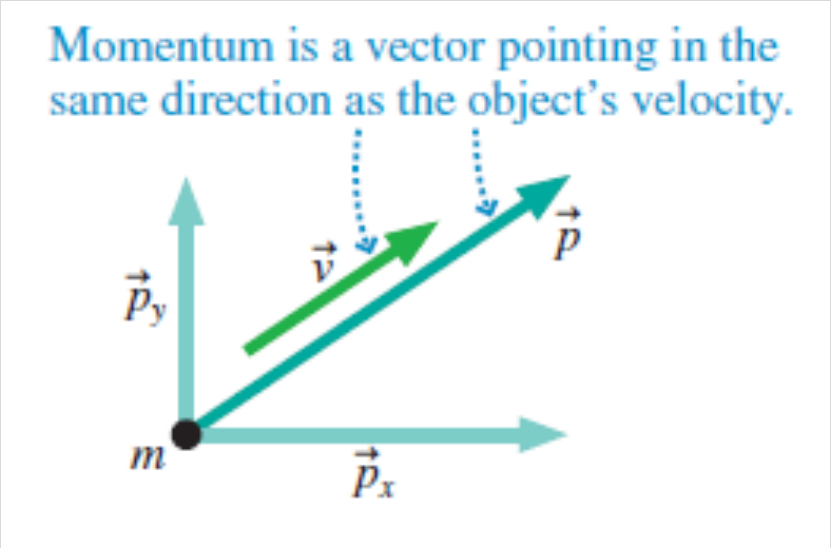
$\Rightarrow mv_f - mv_i = p_f - p_i = \Delta p$

(p: momentum)

Momentum
Definition

To make sense of the previous equation for Impulsive forces, we need to define a new quantity called **Momentum**.

$$\vec{p} = m\,\vec{v}$$



Unit: $kg\ m/s$.

Momentum
Definition

Newton's 2nd Law in terms of Momentum as:

$$\vec{F} = m\vec{a} = \cdots m\frac{d\vec{v}}{dt} = \frac{d(m\vec{v})}{dt}$$

$\vec{F} = \frac{d\vec{p}}{dt}$ can apply when mass not const

$$F = \frac{d\vec{p}}{dt} = \frac{d(m\vec{v})}{dt} = m\frac{d\vec{v}}{dt} + \vec{v}\frac{dm}{dt} = m\vec{a} + 0$$

Product: $(uv)' = uv' + vu'$

Forces can be defined as the rate of change of momentum over time.

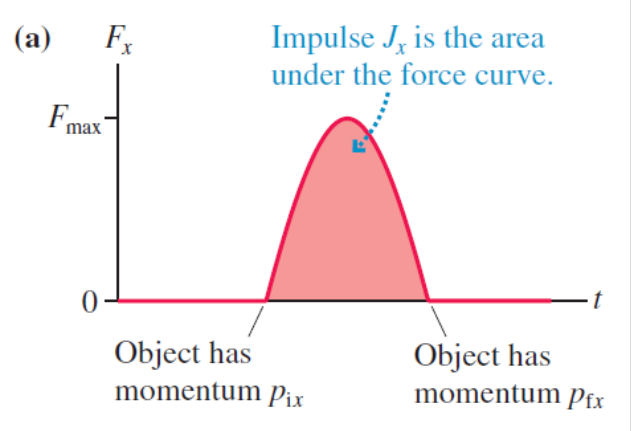
It is more general than our earlier version $\vec{F} = m\vec{a}$ → allows for the possibility that the mass of the object might change, such as a rocket that is losing mass as its burns fuel.

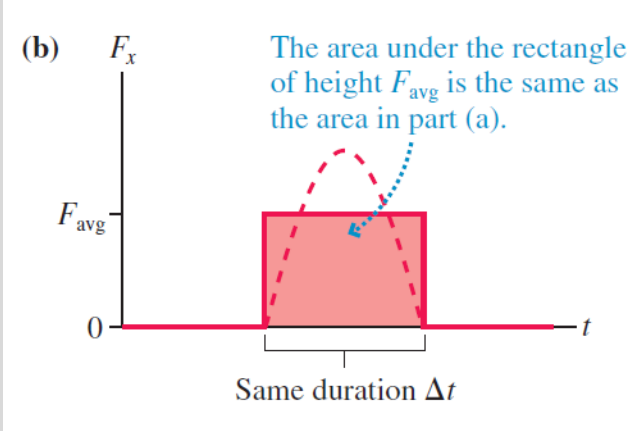
Impulse

Momentum principle

$$\Delta p = \int_{t_i}^{t_f} F(t)\, dt = J$$

→ time integral of force → **impulse**

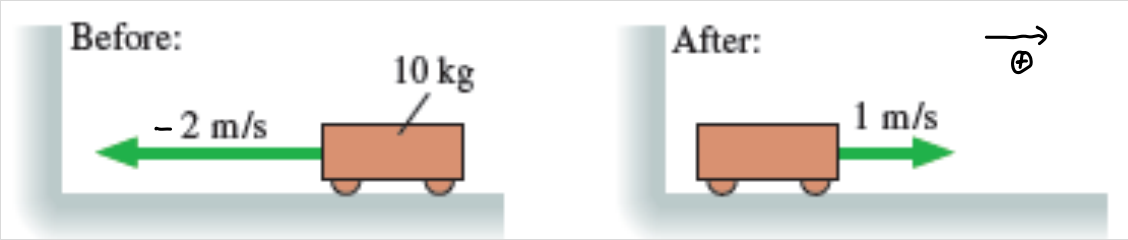




Momentum principle: An impulse delivered to an object causes the object's momentum to change.

Momentum
Examples

Example 1: A 10 kg cart is moving at a constant speed of 2 m/s toward a wall. After it collides with the wall, it can be observed to be moving away from the wall at speed 1 m/s. What was its change in momentum?

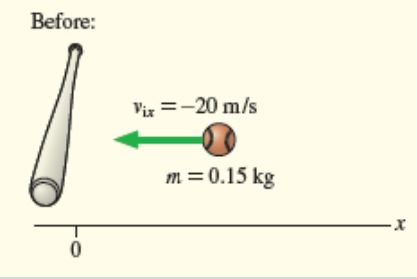


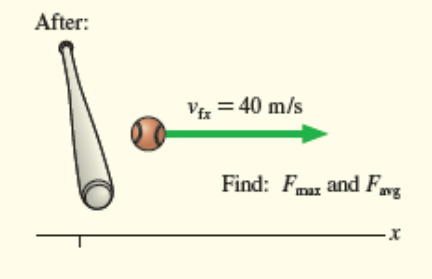
$\Delta P = mv_f - mv_i = 10(1-(-2)) = 10(3) = 30\ kg\ m/s$ N/s

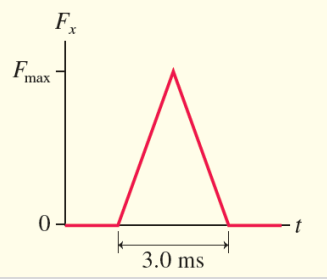
W → change of energy ΔKE
impulse → change of momentum
→ $\Delta KE = W = \int f\,dx$
$\Delta p = J = \int F\,dt$

Momentum
Examples

Example 2: A 150 grams baseball is thrown toward a batter. Just before the ball it hits the bat, its speed is measured at 20 m/s. After the batter hits the ball, its velocity is measured at 40 m/s. High speed camera shows that the collision happens within 3.0 ms. What maximum force and the average force of the bat on the ball? The interaction force between the bat and the ball is plotted below.







$\Delta P = \int F\,dt \Rightarrow m(v_f - v_i) = \frac{1}{2} F_{max} \cdot \Delta t \quad \therefore F_{max} = \frac{2m(v_f - v_i)}{dt} \Rightarrow 6000\,N$

$\frac{1}{2} F_{max}\,\Delta t = F_{avg}\,\Delta t$

$\therefore F_{avg} = F_{max}/2$

Momentum
Examples

Stop to think: A 10 g rubber ball and a 10 g clay ball are thrown at a wall with equal speeds. The rubber ball bounces with the same speed as before, the clay ball sticks. Which ball delivers a large impulse to the wall?

$$\Delta p = m\left[v - (-v)\right] = 2mv \quad \text{(rubber)}$$

$$\Delta p = m\left[0 - (-v)\right] = mv \quad \text{(clay)}$$

a. The clay ball delivers a larger impulse because it sticks.
b. The rubber ball delivers a larger impulse because it bounces.
c. They deliver equal impulses because they have equal momenta.
d. Neither delivers an impulse to the wall because the wall doesn't move.

2. Conservation of Momentum

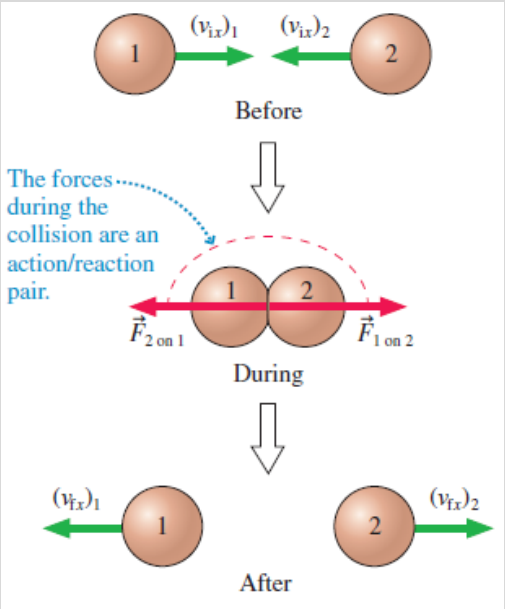
- Involking Newton's 3rd Law
- Complex collision problems can be solved without knowing much about the forces involved during contact

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Conservation of Momentum
Introduction

Newton's 2nd law → MOMENTUM PRINCIPLE.

Now, involking Newton's 3rd law → CONSERVATION of MOMENTUM



$$p_1 + p_2 = const$$

Ball1: $\Delta \vec{p}_1 = \int \vec{F}_{2\text{ on }1}(t)\,dt$

Ball2: $\Delta \vec{p}_2 = \int \vec{F}_{1\text{ on }2}(t)\,dt$

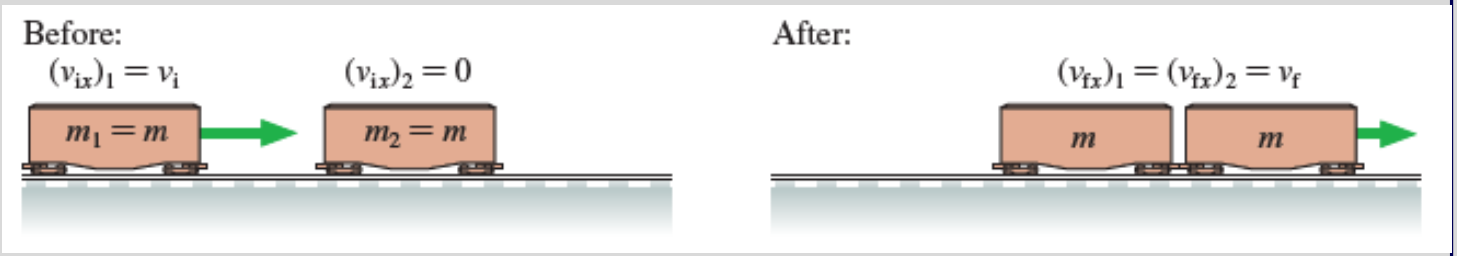
$\vec{F}_{2\text{ on }1} = -\vec{F}_{1\text{ on }2} \Rightarrow \Delta\vec{p}_1 + \Delta\vec{p}_2 = \int (\vec{F}_{2\text{ on }1} + \vec{F}_{1\text{ on }2})\,dt$ ($\Delta\vec{p}$; 0)

$\therefore \Delta\vec{p} = 0$ or $\vec{p}_1 + \vec{p}_2 = const$

Conservation of Momentum

Example

Example 1: car 1 is moving with velocity v_i while car 2 is at rest. Afterward, they roll together with the common final velocity v_f. Furthermore, $m_1 = m_2 = m$. Determine the final velocity v_f.



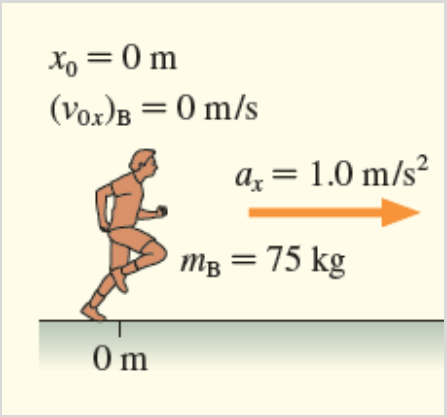
$\Delta P = 0 \Rightarrow \underbrace{\vec{P}_{1i} + \vec{P}_{2i}}_{\text{init momentum}} = \underbrace{P_{1f} + P_{2f}}_{\text{final momentum}} \Rightarrow mv_i + m0 = 2m v_f \Rightarrow v_f = \frac{v_i}{2}$

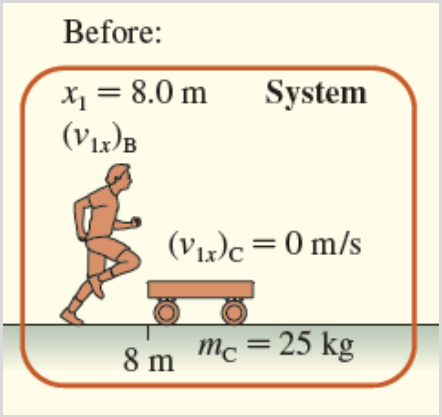
Remark: We can predict the final speed without knowing anything about the complex interaction between the two cars as they collide.

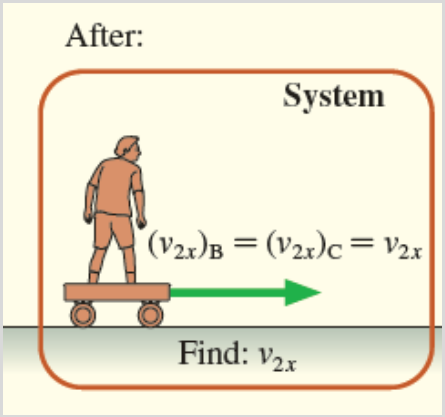
Conservation of Momentum

Example

Example 2: Bob sees a stationary cart 8.0 m in front of him. He decides to run to the cart as fast as he can, jump on, and roll down the street. Bob has a mass of 75 kg and the cart's mass is 25 kg. If Bob accelerates at a steady $1.0\ m/s^2$, what is the cart speed just after Bob jumps on?







3. Collision

An application of momentum

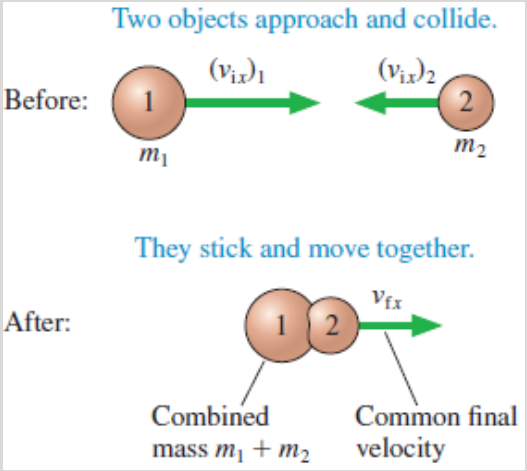
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Collision

Two types

Inelastic collision

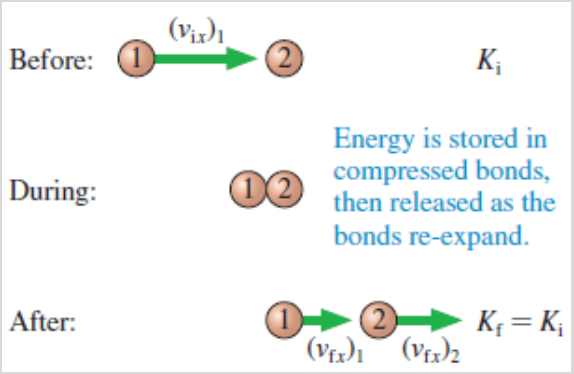
Perfectly Inelastic collision is where an object **stick and move together**.



Example: A bullet shooting at an object and embedded itself in the object.

Elastic collision

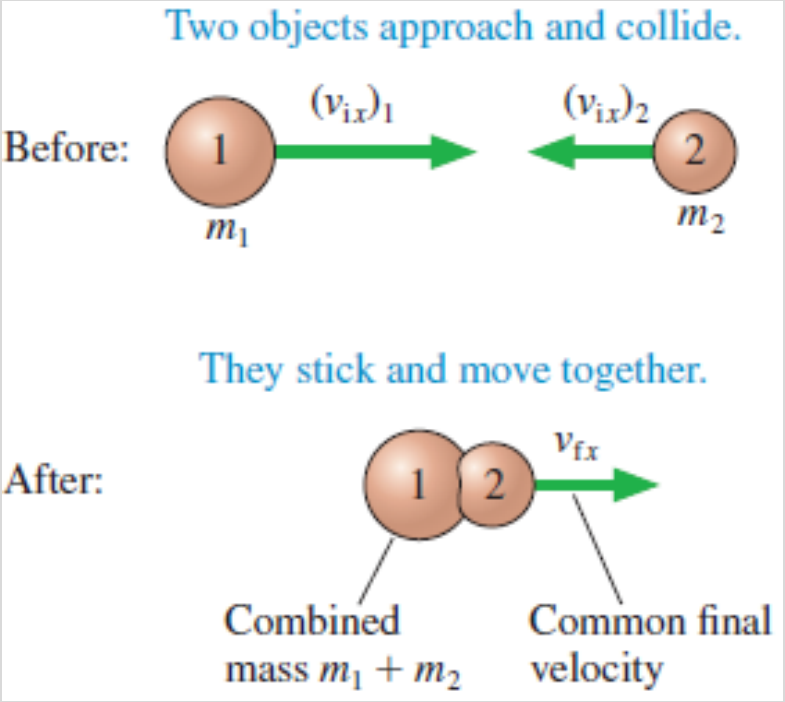
Perfectly Elastic Collision is where the objects **bounce apart with no loss of energy**.



Example: a billiard ball hitting another billiard ball and bounce away.

Collision
Inelastic collision

Perfectly Inelastic collision is where an object **stick and move together**.



$$\Delta P = \vec{P}_1 + \vec{P}_2 = \text{const}$$

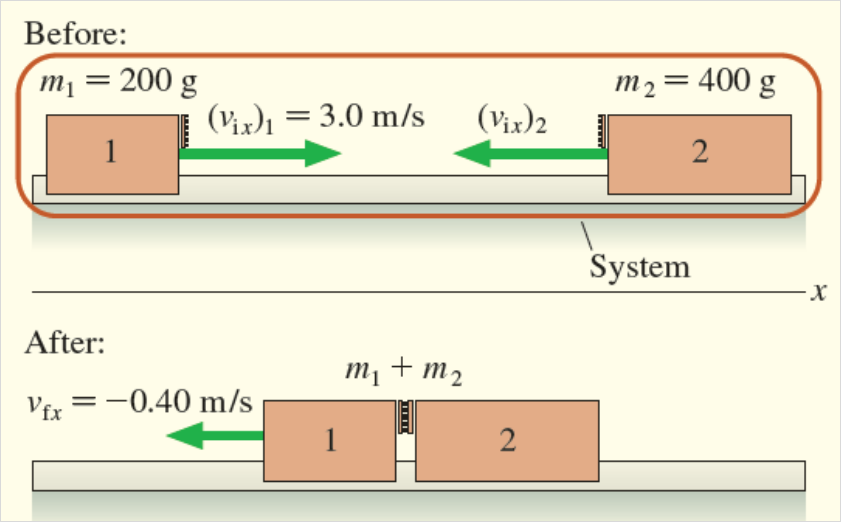
$$\Rightarrow = \vec{P}_{1i} + \vec{P}_{2i} = \vec{P}_f$$

$$m_1 v_{i1} + m_2 v_{i2} = (m_1 + m_2) V_f$$

Collision

Inelastic collision

Example: What was the initial speed of the second cart? $v_{i2} = ?$

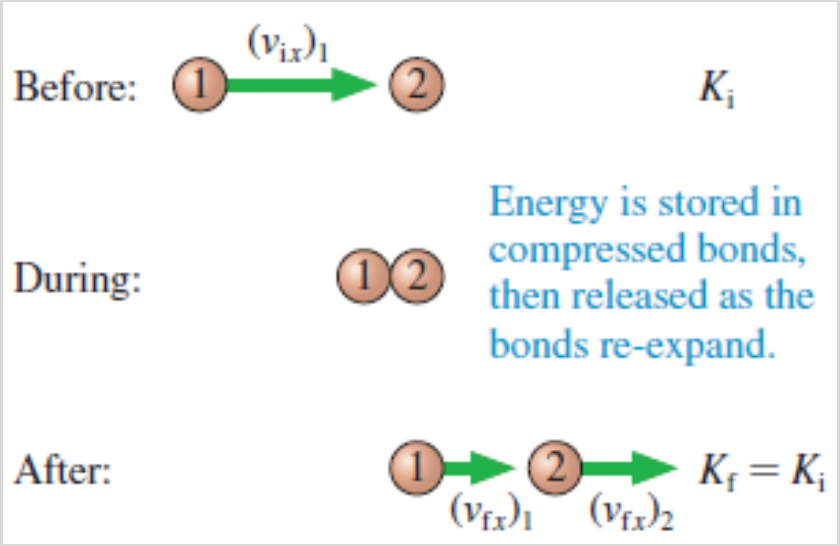


$$m_1 v_{i1} + m_2 v_{i2} = (m_1 + m_2) v_f$$

$$\Rightarrow v_{i2} = v_f + \frac{m_1}{m_2} (v_f - v_{i1})$$

Collision
Inelastic collision

Perfectly Elastic Collision is where the objects **bounce apart with no loss of energy**.



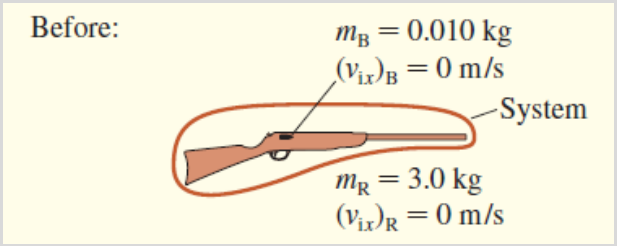
$\Delta P = 0 \Rightarrow m_1 v_{i1} + m_2 v_{i2} = m_1 v_{f1} + m_2 v_{f2}$

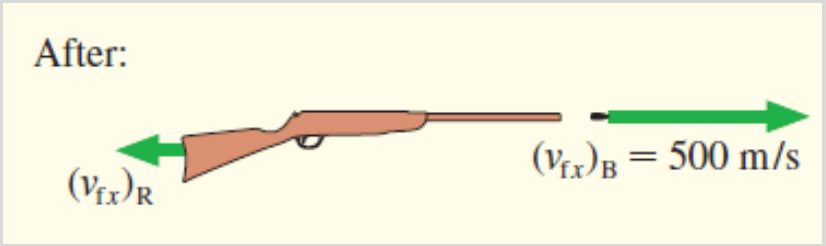
$\Delta KE = 0$

$\Rightarrow \frac{1}{2} m_1 v_{i1}^2 + \frac{1}{2} m_2 v_{i2}^2 = \frac{1}{2} m_1 v_{f1}^2 + \frac{1}{2} m_2 v_{f2}^2$

Explosion

Example: A 10 g bullet is fired from a 3.0 kg rifle with a speed of 500 m/s. What is recoil speed of the rifle?





$\Delta P = 0 \Rightarrow P_i = P_f = m_R v_{fR} + m_B v_{fB} = 0$ ($P_i = 0$)

$$v_{fB} = \frac{-m_B v_{fB}}{m_R} = -1.7 \text{ m/s}$$

4. Momentum in Two Dimensions

Collisions and Explosions in 2D

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Momentum in 2D

Many interesting examples of collisions and explosions involve motion in a plane.



Collisions and explosions often involve motion in two dimensions.

$$(p_{fx})_1 + (p_{fx})_2 + (p_{fx})_3 + \cdots = (p_{ix})_1 + (p_{ix})_2 + (p_{ix})_3 + \cdots$$ in x-axis

$$(p_{fy})_1 + (p_{fy})_2 + (p_{fy})_3 + \cdots = (p_{iy})_1 + (p_{iy})_2 + (p_{iy})_3 + \cdots$$ in y-axis